Surrogate models for Monte Carlo uncertainty propagation with few runs of an expensive model: specification, selection and a valid surrogate contribution

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Abstract

When the measurement model is an expensive simulation, Monte Carlo propagation of distributions according to GUM Supplement 1 is carried out on a surrogate fitted to a few tens of model runs, and the surrogate's error becomes part of the uncertainty budget. We compare Gaussian processes (GPs) of different specifications, sparse polynomial chaos expansions, committees of surrogates, and space-filling, adaptive and threshold-targeted designs on twelve models with up to 20 inputs, including GUM-type input distributions and a computational fluid dynamics (CFD) simulation of a gas release behind a building, with 10 to 80 runs and 30 paired repetitions against reference Monte Carlo evaluations. The specification of the surrogate mattered more than the choice of strategy: an estimated kernel amplitude, a quadratic trend and the log scale for positive quantities reduced the error by factors of up to 18, 400 and 9, and choosing among such candidates by their leave-one-out error recovered practically all of the gain without access to the true model. Committees did not beat a GP with a trend, and variance-driven adaptive designs gave no consistent gain at 20 runs and helped only with larger budgets on smooth models, whereas a design targeted at a threshold in the tail reduced the error of exceedance probabilities up to ninefold. Adding the leave-one-out residuals of the surrogate to every Monte Carlo trial brought the achieved coverage of the GUM S1 95% coverage interval to 91–97% on all models, and jackknife+ prediction intervals covered 95–100% of test points, whereas the GP's own predictive intervals covered as few as 68%, and its intervals for a 95th percentile contained the true value in a median of only 63% of repetitions. The results are condensed into a validated procedure for surrogate-based uncertainty evaluation.

Keywords: measurement uncertainty; GUM Supplement 1; Monte Carlo method; surrogate model; Gaussian process; polynomial chaos; conformal prediction; gas dispersion.

1 Introduction

Evaluating measurement uncertainty by the propagation of distributions, following Supplement 1 to the Guide to the Expression of Uncertainty in Measurement (GUM S1) [1, 2], requires the measurement model $Y = f(\mathbf{X})$ to be evaluated once per Monte Carlo trial, typically 10^5 to 10^6 times [3, 4]. For a growing class of quantities f is a numerical simulation. A computational fluid dynamics (CFD) model of an accidental ammonia release in a factory [5] can take hours to days per run, so that a practical budget is a few tens of runs, three or four orders of magnitude short of what the Monte Carlo method needs. The same gap arises in scatterometry, where it has been studied in detail [6], and in many inverse and model-based measurements.

The usual remedy is a *surrogate* (emulator, metamodel) fitted to the available runs and evaluated in place of f [7, 8]. Two questions then decide whether the resulting uncertainty statement can be trusted: which surrogate reproduces the reported quantities best for a given number of runs, and how the surrogate's own error is to be included in the uncertainty budget, since it is not negligible at such budgets. This paper answers both with a controlled comparison on twelve models, including a CFD dispersion case, and validates the answers against the true models.

1.1 Related work

Gaussian process emulators. Gaussian process (GP) regression, or kriging, is the standard surrogate for deterministic computer experiments [9, 10]. The Bayesian treatment of O'Hagan and co-workers [7, 11] reads the GP posterior as the uncertainty about f itself, which suggests using the posterior variance as the surrogate contribution to an uncertainty budget. A common guideline asks for about ten runs per input [12], and the effect of modelling choices such as the correlation family, the mean function and the treatment of the hyperparameters on GP accuracy has been assessed systematically [13].

Polynomial chaos. Polynomial chaos expansions (PCE) represent Y in polynomials orthonormal with respect to $g_{\mathbf{X}}$ [14], from which the moments and sensitivity indices follow directly [15]. With few runs, sparse expansions selected by least-angle regression [16] and a corrected leave-one-out error are the method of choice [17]. Direct comparisons of PCE and GP emulators on expensive simulators found neither uniformly better [18].

Ensembles and committees. Combining several surrogates with weights derived from their cross-validation errors protects against choosing a poor surrogate [19, 20]; the spread of the members has also been used as an interval for the output, in the spirit of probability boxes [21].

Adaptive designs. Instead of fixing all runs in advance, runs can be added where the surrogate is most uncertain, by its posterior variance [22], by the disagreement of a committee, or by measures such as the spectral uncertainty of Lin et al. [23]. For the probability of exceeding a threshold, the AK-MCS method of Echard et al. [24] places runs where the sign of $f - T$ is most uncertain.

Surrogates in metrology. In scatterometry, surrogates have been compared for the Bayesian inverse problem [25], used to make Bayesian shape reconstruction of lithography masks feasible [26], and sampling methods for expensive models have been surveyed [6]. These works concern the inverse problem or the efficiency of sampling; the present study concerns the forward propagation of GUM S1 and how the surrogate error enters the uncertainty budget.

Validity of the surrogate contribution. Whether any of these uncertainty descriptions has the coverage it claims is rarely checked against the true model. Conformal prediction [27], and in particular the jackknife+ [28], gives prediction intervals with a coverage guarantee that holds for any fitted model, using only leave-one-out fits. To our knowledge it has not been compared with GP posterior intervals as the surrogate contribution in a GUM S1 evaluation.

Benchmarks. Comparisons of surrogates are sensitive to the test functions and to the number of repetitions. We use standard functions from the literature [29, 30], GUM-type input distributions, a physical dispersion formula and a CFD case, with 30 paired repetitions and statistics adjusted for multiple comparisons.

1.2 Earlier analysis and contributions

An earlier analysis by the first author combined a heterogeneous committee, the spectral acquisition criterion and dynamic model management, and reported that a passively sampled GP fails catastrophically

on large input domains, a failure the committee avoided. The present study re-examines these claims with correctly specified baselines and extends them to the general question above. Its contributions are:

1. a paired comparison, on twelve models with up to 20 inputs, uniform and GUM-type input distributions and a CFD dispersion case, of GP specifications, sparse PCE, committees and adaptive and targeted designs at budgets of 10 to 80 runs, scored on the quantities a GUM S1 evaluation reports;
2. the finding that the GP failure of the earlier analysis is a kernel-specification artefact, and that specification (amplitude, anisotropy, trend) and variable transformation matter more than the choice between single surrogates, committees and adaptive designs;
3. a validation, against the true models, of three ways to include the surrogate error in GUM S1: the GP posterior, a cross-validation (CV) normal error and resampled leave-one-out residuals, with jackknife+ intervals as a distribution-free reference;
4. a procedure for surrogate-based GUM S1 evaluations (Fig. 6), and code, per-seed records and an audit script that recompute every number reported [31].

2 Surrogate-based propagation and the surrogate contribution

2.1 Propagation of distributions through a surrogate

Let $Y = f(\mathbf{X})$ be the measurement model, $g_{\mathbf{X}}$ the joint probability density of the inputs (here with independent components) and $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, $y_i = f(\mathbf{x}_i)$, the N available model runs. A surrogate $\hat{f}$ is fitted to the runs and the Monte Carlo method of GUM S1 [2] is applied to $\hat{f}$: M draws $\mathbf{x}^{(r)} \sim g_{\mathbf{X}}$ give $\hat{y}^{(r)} = \hat{f}(\mathbf{x}^{(r)})$, from which the estimate y (the mean), the standard uncertainty $u(y)$ (the standard deviation), the probabilistically symmetric 95% coverage interval $[y_{0.025}, y_{0.975}]$, the 95th percentile $y_{0.95}$ and the probability $P(Y > T)$ of exceeding a threshold T are computed. We call these the *reported quantities*.

Surrogates are fitted on a *latent* scale $z = \varphi(y)$, with $\varphi = \log$ for the log-variable variants and the identity otherwise, and predictions are mapped back by φ^{-1}. For a Gaussian latent predictive distribution, exp of the latent mean is the predictive median of Y; no bias correction is applied.

2.2 The surrogate error as an uncertainty contribution

The surrogate error $e(\mathbf{x}) = \varphi(f(\mathbf{x})) - \varphi(\hat{f}(\mathbf{x}))$ is unknown. Following Kennedy and O'Hagan [11], who treat model inadequacy as an uncertainty source, we propagate

$$Y = \varphi^{-1}\Big(\varphi(\hat{f}(\mathbf{X})) + E\Big), \tag{1}$$

with E a random variable describing the surrogate error, drawn independently for every Monte Carlo trial. Three models for E are compared, all available from the N runs alone:

- *GP*: $E \mid \mathbf{x} \sim \mathrm{N}(0, s^2(\mathbf{x}))$, with $s(\mathbf{x})$ the predictive SD of a Gaussian process surrogate;
- *CV-normal*: $E \sim \mathrm{N}(0, s^2_{\mathrm{LOO}})$, with $s^2_{\mathrm{LOO}} = N^{-1} \sum_i r_i^2$ and $r_i = \varphi(y_i) - \varphi(\hat{f}_{-i}(\mathbf{x}_i))$ the leave-one-out (LOO) residuals; $\hat{f}_{-i}$ is the surrogate re-fitted, hyperparameters included, without run i;
- *resampling*: E drawn with equal probabilities from $\{r_1, \ldots, r_N\}$.

The same ingredients give pointwise 95% prediction intervals for $f(\mathbf{x})$: the GP interval $\hat{z} \pm 1.96\, s(\mathbf{x})$, the CV-normal interval $\hat{z} \pm 1.96\, s_{\mathrm{LOO}}$, and the jackknife+ interval [28]

$$\left[\, q^{-}_{\alpha}\{\hat{z}_{-i}(\mathbf{x}) - |r_i|\},\ q^{+}_{\alpha}\{\hat{z}_{-i}(\mathbf{x}) + |r_i|\} \,\right], \tag{2}$$

where q^{-}_{α} is the $\lfloor \alpha(N+1) \rfloor$-th smallest and q^{+}_{α} the $\lceil (1-\alpha)(N+1) \rceil$-th smallest of the N values, $\alpha = 0.05$. For exchangeable runs the jackknife+ interval covers f with probability at least $1 - 2\alpha$ whatever the surrogate [28]; it is a member of the conformal family [27].

Why resampling the LOO residuals can be expected to work. For an input $\mathbf{X} \sim g_{\mathbf{X}}$ drawn independently of the runs, $\varphi(f(\mathbf{X})) = \varphi(\hat{f}(\mathbf{X})) + e(\mathbf{X})$ exactly, so equation (1) gives the true distribution of the output if E behaves as $e(\mathbf{X})$. If the run inputs and $\mathbf{X}$ are exchangeable, as for independent draws from $g_{\mathbf{X}}$, each r_i is distributed as the error of a surrogate fitted to $N-1$ runs at a new input, so the empirical distribution of the r_i estimates the distribution of $e(\mathbf{X})$, on the conservative side because $N-1$ rather than N runs are used [32]. Three approximations remain. Resampling draws E independently of $\mathbf{X}$: it reproduces the variance of the error but drops its covariance with $\varphi(\hat{f}(\mathbf{X}))$, which is zero in expectation when $\hat{f}$ is the posterior mean of a correctly specified model but not in general. A Latin hypercube design is close to, not exactly, exchangeable with independent draws. And a feature of f that no run reaches leaves no trace in any r_i (Section 5.8). At a fixed input $\mathbf{x}$, the same construction with $\hat{f}_{-i}(\mathbf{x})$ in place of $\hat{f}(\mathbf{x})$ is the predictive distribution behind the jackknife+ interval, one of the conformal predictive distributions, which are valid in finite samples under exchangeability [28, 33]. For the distribution of Y, which mixes over the inputs, no such guarantee is available, and the coverage study of Section 5.7 is its empirical check.

For a reported quantity θ itself (the mean, SD, $y_{0.95}$ or $P(Y > T)$ of the output over a fixed set of input draws), two 95% intervals are compared: the 2.5th and 97.5th percentiles of θ over joint sample paths of the GP posterior, and the jackknife interval $\hat{\theta} \pm 1.96\, \mathrm{SE_J}$ with $\mathrm{SE_J^2} = \frac{N-1}{N} \sum_i (\hat{\theta}_{-i} - \bar{\theta})^2$, where $\hat{\theta}_{-i}$ is computed from $\hat{f}_{-i}$.

Hyperparameter uncertainty. The GP intervals above plug in the maximum-likelihood (ML) hyperparameters, which may make them too narrow with few runs. Three variants test whether the GP's shortfall is a plug-in effect: (i) a Student-t process, with the overall variance scale and a constant mean integrated out under the reference prior $p(\beta, \sigma^2) \propto \sigma^{-2}$ and the correlation hyperparameters at their ML values (the standard Bayesian emulator [7], $N-1$ degrees of freedom); (ii) a fully Bayesian GP, all hyperparameters integrated out by Markov chain Monte Carlo (affine-invariant ensemble sampler [34], uniform prior on each log-hyperparameter within the ML search range, 800 steps of which 300 burn-in, 200 posterior draws; $N = 20$ only, for cost); and (iii) the GP SD recalibrated by the LOO standardised residuals $z_i = r_i / s_{-i}(\mathbf{x}_i)$: pointwise $\hat{z} \pm q\, s(\mathbf{x})$ with q the $\lceil 0.95(N+1) \rceil$-th smallest $|z_i|$, $E = s(\mathbf{x})\, z_J$ in equation (1), and sample paths with the covariance scaled by the mean of z_i^2. The last is a hybrid that keeps the GP's localisation of the error and takes its scale from the runs. Each variant is assessed in the same three ways.

3 Test models

Table 1 lists the twelve models. Eight have independent uniform inputs, three have the non-uniform input distributions of a GUM-type evaluation, and one is a CFD simulation. The physics model is the benchmark of the earlier analysis; its quadratic term carries practically all the output variance on $[-200, 200]^3$ and lies in the class of a degree-2 polynomial. The plume model is the ground-level

concentration of a continuous ammonia release from a Gaussian plume with ground reflection,

$$C = \frac{10^3\, Q}{2\pi\, u\, \sigma_y \sigma_z} \exp\!\left(-\frac{y^2}{2\sigma_y^2}\right) 2\exp\!\left(-\frac{H^2}{2\sigma_z^2}\right) \ \mathrm{mg\,m^{-3}}, \tag{3}$$

with Briggs rural coefficients for neutral stability, $\sigma_y = 0.08x(1 + 10^{-4}x)^{-1/2}$ and $\sigma_z = 0.06x(1 + 1.5 \times 10^{-3}x)^{-1/2}$. The borehole [29, 35, 36], Ishigami [29], OTL circuit and piston [37] and wing weight [38] functions are standard test functions for computer experiments. To test whether the results hold when the inputs outnumber what 20 runs can resolve, the OTL circuit is also run with 14 additional inputs that do not affect the output (uniform on $[100, 200]$, as in the screening study of Moon [39]), so that $d = 20$ and $N = 20$ is one run per input. The flood model [40] gives the maximal annual underflow of a river with truncated Gumbel, truncated normal, triangular and uniform inputs. The borehole model is also run with the input model of Harper and Gupta [35] (normal borehole radius, lognormal radius of influence) and the plume model with GUM-type inputs: lognormal emission rate (median $500\,\mathrm{g\,s^{-1}}$, geometric SD 2), a Weibull wind speed (shape 2, scale $4\,\mathrm{m\,s^{-1}}$, truncated to $[0.5, 15]\,\mathrm{m\,s^{-1}}$) and a normal crosswind offset (SD 15 m). All formulas and input models were checked against the independent implementation in UQTestFuns [30] (relative differences below 10^{-10} for outputs and 10^{-6} for marginal quantiles).

A further analytical model, reported separately, tests the limits of the procedure: a smooth function of four uniform inputs on $[0, 1]^4$ plus a narrow Gaussian bump, $f = x_1 + x_2^2 + 0.5\sin(2\pi x_3) + 0.5x_4 + 3.6\exp(-\|\mathbf{x} - 0.7\|^2/(2 \cdot 0.13^2))$. The bump carries about 10% of the output variance in a region (bump above 10% of its height) of 3% of the input volume, which a Latin hypercube of 20 runs misses with probability 0.53 and one of 40 runs with probability 0.26 (Section 5.8). Two variants test assumptions of the main study. In the first, the inputs are dependent: the borehole model with GUM-type inputs, with a Gaussian copula of correlation 0.5 between the two transmissivities and between the two heads. Designs are drawn in the space of independent uniform variables and mapped to the inputs by the inverse Rosenblatt transformation of the copula, so the surrogates and the propagation are otherwise unchanged. In the second, the output is a vector: the plume model at three receptors on the plume axis, at x, $x + 200$ m and $x + 400$ m, gives three strongly correlated output quantities, propagated as in GUM S2 [41]. Each component has its own log-scale GP, and the surrogate error is added on the log scale in one of three ways: as the LOO residual vector of one run, drawn with equal probabilities (*joint* resampling, which keeps the correlation between the components), as residuals drawn independently for each component, or as independent normal errors with the LOO RMS error of each component. The coverage region is the hyperellipsoidal region of GUM S2, from the mean and covariance matrix of the propagated sample with the coverage factor set so that 95% of the sample lies inside, computed on the original and on the log scale. Its achieved coverage is evaluated under 2×10^5 draws of the true model, and its volume is compared with that of the same region computed from the true model.

Exceedance is assessed at a fixed threshold where the model has one (the IDLH level for ammonia, $210\,\mathrm{mg\,m^{-3}}$, about 300 ppm, for the plume and CFD models), and otherwise at the reference 95th percentile, for which $P(Y > T) = 0.05$ by construction. Flooding has a probability below 10^{-5} for the flood input model, so its threshold is the reference 5th percentile of the underflow, $P(Y < T) = 0.05$.

3.1 CFD dispersion case

The CFD case replaces the plume formula by a Reynolds-averaged Navier–Stokes (RANS) simulation of a continuous ground-level release upwind of a building, solved with OpenFOAM v1912 [42]. The domain is two-dimensional (a vertical plane along the wind; the building and the release are infinitely wide), from 80 m upstream to 250 m downstream of the windward face and 150 m high. The building

Table 1: Test models. d: number of inputs; U: independent uniform inputs. The models below the line test the limits and assumptions of the procedure and are reported separately.

Model	d	Inputs	Character
Physics	3	$\mathrm{U}[-200, 200]^3$, noise SD 0.1	quadratic bowl plus small oscillatory terms
Plume	5	U: Q, u, H, x, y	multiplicative, positive, heavy right tail
Borehole	8	U, ranges of [29]	smooth, multiplicative, positive
Ishigami	3	$\mathrm{U}[-\pi, \pi]^3$	strongly non-linear, interactions
OTL circuit	6	U [37]	smooth, nearly additive, positive
Piston	7	U [37]	non-linear, positive
Wing weight	10	U [38]	smooth, multiplicative, positive
OTL circuit, 20 inputs	20	U, 14 inactive [39]	as OTL circuit; sparse
Borehole (GUM)	8	normal, lognormal, uniform [35]	as borehole; input tails
Plume (GUM)	5	lognormal, truncated Weibull, uniform, normal	as plume; heavier input and output tails
Flood	8	truncated Gumbel and normal, triangular, uniform [40]	smooth, lower-tail threshold
CFD dispersion	6	U: U_{ref}, z_0, H_{b}, Sc_t, x_r, Q	RANS solution: wake recirculation, numerical error
Bump (failure mode)	4	$\mathrm{U}[0, 1]^4$	smooth part plus a narrow bump that the runs can miss
Borehole, dependent	8	as Borehole (GUM), Gaussian copula	dependent inputs
Plume, 3 receptors	5	as Plume	vector output (GUM S2)

is 10 m deep and H_{b} high; the release is a unit line source 20 m upstream of the building, 0.5 m above the ground. The inflow is the neutral atmospheric surface layer of Richards and Hoxey [43], $U(z) = (u_*/\kappa)\ln((z + z_0)/z_0)$, $k = u_*^2/\sqrt{C_\mu}$, $\varepsilon = u_*^3/(\kappa(z + z_0))$, scaled to the wind speed U_{ref} at 10 m, with the standard k–ε model ($\sigma_\varepsilon = 1.11$ for consistency with the inflow), a rough-wall function on the ground [44], smooth building walls and a slip top boundary. The concentration s of the passive scalar solves $\nabla \cdot (\mathbf{U}s) - \nabla \cdot ((\nu/Sc + \nu_t/Sc_t)\nabla s) = S$, with $Sc = 0.7$ and the turbulent Schmidt number Sc_t as an input, since its appropriate value is uncertain and varies between flows [45]. Flow and scalar are solved together by the SIMPLE algorithm on 7776 hexahedral cells, refined towards the building, until all scaled residuals fall below 10^{-5} (pressure) and 10^{-6} (other fields), or 4000 iterations, typically in 1000–1500 iterations (about 20 s on one core). The output is the concentration at 1.5 m height a distance x_r downstream of the leeward face, $C = 10^3\, Q\, s(x_r)\ \mathrm{mg\,m^{-3}}$ for a release rate Q per metre of crosswind width; since the scalar is passive, x_r and Q need no separate run. The six inputs are uniform: $U_{\mathrm{ref}} \in [1, 6]\ \mathrm{m\,s^{-1}}$, $z_0 \in [0.01, 0.3]$ m, $H_{\mathrm{b}} \in [6, 15]$ m, $Sc_t \in [0.5, 1.3]$, $x_r \in [5, 120]$ m and $Q \in [1, 20]\ \mathrm{g\,s^{-1}\,m^{-1}}$.

The case contains what the plume formula lacks: flow separation at the building edges, a recirculating wake whose length and strength change with H_{b} and z_0, receptors inside and beyond the wake, and a discretisation and iteration error that varies with the inputs. It is a benchmark, not a validated prediction of a particular site: the model is *defined* as this discretised RANS solution, which is what a surrogate of an expensive simulation has to reproduce. A reference Monte Carlo of 2×10^5 CFD runs is not affordable. The reference distribution is built instead from 800 runs at independent uniform draws of the four flow inputs, each combined with 50 independent draws of (x_r, Q): 40 000 reference points, of which 4000 (five per run) form the test set and 1600 (two per run) the input set for derived quantities. The grid sensitivity of the case is reported in Section 5.9.

4 Surrogates, designs and protocol

4.1 Surrogates

Gaussian processes. All GPs are fitted on standardised inputs and outputs by maximising the log marginal likelihood (L-BFGS-B, five random restarts) with scikit-learn. The specifications, in order of increasing completeness, are: (i) the *original* kernel of the earlier analysis, a squared exponential with one length scale and an amplitude *fixed* at 1 on the standardised output; (ii) with the amplitude estimated; (iii) with one length scale per input (automatic relevance determination, ARD) but the amplitude fixed; (iv) with both; (v) a Matérn-5/2 ARD kernel with amplitude and a nugget; (vi) (v) plus an amplitude-scaled quadratic polynomial kernel, i.e. a GP with a quadratic trend whose coefficients are integrated out. Length scales are bounded to $[10^{-2}, 10^{3}]$, the amplitude to $[10^{-3}, 10^{3}]$ and the nugget variance to $[10^{-8}, 10^{-1}]$. Variants (v) and (vi) are also fitted on the log scale for positive outputs, with logarithmic transformation of inputs whose support is positive.

Sparse polynomial chaos. A sparse PCE [17] with an orthonormal Legendre basis on inputs mapped to $[-1, 1]$ (affinely for uniform inputs, through their distribution functions otherwise), candidate sets of total degree $p = 1, \ldots, 5$ with hyperbolic truncation ($q = 1$ for $p \leq 2$, $q = 0.75$ above), least-angle regression [16] to order the basis terms, ordinary least squares on every active set along the path, and selection of the active set and degree by the corrected LOO error [17]. It is fitted on the raw and on the log scale.

Committees. Nine members: two GPs (squared exponential and Matérn-3/2, both ARD with amplitude), PCEs [14, 15] of degrees 2 and 3 (Bayesian ridge regression), random forests of 50 and 100 trees, a support vector regressor and two neural networks. Members are combined uniformly or with weights $w_k \propto [\mathrm{MSE}_{\mathrm{CV},k}(1+\pi_k)]^{-1}$, with $\mathrm{MSE}_{\mathrm{CV},k}$ the three-fold cross-validation error of member k and $\pi_k \geq 0$ a penalty for predictions below a known physical bound [19, 20].

Stacking. As an alternative to choosing one candidate surrogate (Section 5.5), the candidates are combined by stacking [46]: non-negative weights that minimise the squared error of the weighted sum of their LOO predictions (non-negative least squares) are applied to the candidates fitted to all runs.

4.2 Designs

Passive designs are N-point Latin hypercubes drawn on $[0, 1]^d$ and mapped to the inputs by the inverse distribution functions. Adaptive designs start from N_0 Latin hypercube runs ($N_0 = 5$ for $N \leq 20$, $N/4$ above) and add one run at a time, refitting after every run: *GP-SD* adds the candidate with the largest posterior SD [22]; the adaptive committees add the candidate where the active members disagree most, by the variance of their predictions or by the spectral measure UEigV [23] of the earlier analysis; and *AK-MCS* adds the candidate minimising $|\hat{z}(\mathbf{x}) - \varphi(T)|/s(\mathbf{x})$, the learning function of Echard et al. [24], which concentrates runs near the threshold T of the exceedance probability. *IMSE* adds the candidate that most reduces the integrated posterior variance of the GP [9], averaged over 1000 draws from the input distribution, a criterion aimed at the whole distribution of the output rather than at the largest local uncertainty. Candidates are random draws from the input distribution (300 for GP-SD and IMSE, 2000 for AK-MCS). To test whether the results depend on the type of passive design, the Latin hypercube is replaced, in a separate comparison at $N = 20$, by the maximin Latin hypercube [47] (the one with

Table 2: Median nRMSE over 30 seeds with $N = 20$ runs; bold: best per model. AK-MCS on the log scale for positive outputs. Generated from the result records; 95% bootstrap intervals are in the supplementary summary.

	GP					Sparse PCE		Committee / adaptive			Targeted
Model	orig.	Matérn	+trend	log	log+tr.	raw	log	CV	GP-SD	adapt.	AK-MCS
Physics	0.591	0.079	0.00018	–	–	**0.00014**	–	0.00018	0.070	0.00015	0.084
Plume	0.602	0.597	0.599	0.418	**0.349**	0.725	0.434	0.582	0.681	0.716	0.385
Borehole	0.295	0.089	0.089	**0.011**	0.014	0.364	0.041	0.217	0.082	0.136	0.013
Ishigami	0.956	0.902	**0.884**	–	–	0.906	–	0.889	0.970	0.895	0.999
OTL circuit	0.198	0.070	0.080	**0.037**	0.045	0.118	0.122	0.121	0.064	0.086	0.051
Piston	0.360	0.264	0.287	0.189	0.174	0.439	0.379	0.303	0.246	0.259	**0.106**
Wing weight	0.283	0.121	0.127	0.048	**0.025**	0.176	0.034	0.195	0.128	0.135	0.049
OTL circuit, 20 inputs	0.631	0.103	0.120	**0.049**	0.062	0.257	0.134	0.277	0.112	0.168	0.069
Borehole (GUM inputs)	0.275	0.116	0.117	**0.030**	0.032	0.334	0.309	0.245	0.096	–	0.039
Plume (GUM inputs)	0.696	0.655	0.664	0.365	**0.279**	0.836	0.507	0.664	0.797	–	0.297
Flood	0.227	**0.059**	0.060	–	–	0.210	–	0.109	0.061	–	0.070
CFD dispersion	0.557	0.529	0.500	0.080	**0.060**	0.721	0.208	0.520	0.532	0.448	0.087

the largest smallest distance between two runs among 200 random ones) and by the first N points of a randomly scrambled Sobol' sequence [48].

4.3 Protocol and statistics

Every strategy is run with the 30 seeds 1000–1029 and budgets $N = 10, 20, 40, 80$ (the full set of strategies at $N = 20$). Seeds are shared and the Latin hypercube is seeded from the global random state, so that for a given seed all passive strategies use the identical design and all adaptive strategies the identical start. Accuracy over the input domain is the root mean square error on 5000 test points divided by the reference output SD (normalised RMSE, nRMSE; 1 corresponds to a constant predictor). The surrogates propagate 3×10^4 input draws; the reference uses 2×10^5 draws of the model (Section 3.1 for the CFD case).

For each strategy and model the median over seeds is reported with a 95% percentile-bootstrap confidence interval (10 000 resamples of the seeds). Two strategies are compared on the same seeds by the median of the per-seed ratio of their errors, with a paired bootstrap 95% confidence interval, and by a two-sided Wilcoxon signed-rank test on the log ratios; the p-values of all comparisons of the main study are adjusted together by Holm's method, and those of each further comparison (design types, IMSE, stacking) within that comparison. Coverage of prediction and coverage intervals is the share of test points or of the reference distribution inside the interval, averaged over seeds; coverage of intervals for reported quantities is the share of seeds whose interval contains the true value, with a Wilson 95% interval. The interval score [49] combines width and coverage. Because a practitioner has a single design, coverage is also reported per design, as the share of the 30 designs whose achieved coverage is below 90%.

A comparison without a significant difference does not show that two strategies are equivalent. Such comparisons are classified as equivalent within ±20% (or ±10%) when the 95% bootstrap interval of the median ratio lies inside $[0.8, 1.25]$ (or $[0.9, 1/0.9]$), and as inconclusive otherwise. These margins, the thresholds for coverage (at least 93% pointwise; at least 26 of 30 seeds for a reported quantity, the smallest count not rejected against 95% by a one-sided exact binomial test at the 5% level) and the decisions that the results would trigger were fixed in a research plan committed to the repository [31] before the runs. Finally, the sparse PCE and GP implementations were checked against an independent one, OpenTURNS 1.27 [50], on the 30 designs of six models with uniform inputs.

Table 3: Paired comparisons at $N = 20$: median over 30 seeds of the ratio of the first strategy's error to the second's (nRMSE; for the exceedance rows, the absolute error of $P(Y > T)$). Below 1 favours the first. Bold: Holm-adjusted $p < 0.05$ (two-sided Wilcoxon signed-rank test on log ratios, all comparisons of the study adjusted together). Generated from the result records.

Comparison	Phys.	Plume	Bore.	Ishi.	OTL	Pist.	Wing	OTL20	Bore.G	Plume.G	Flood	CFD
kernel: amplitude + ARD vs original	**0.14**	0.97	**0.31**	0.95	**0.34**	**0.71**	**0.44**	**0.16**	**0.40**	0.93	**0.25**	0.96
log variables vs raw	–	**0.69**	**0.11**	–	**0.49**	**0.75**	**0.40**	**0.45**	**0.26**	**0.54**	–	**0.17**
quadratic trend vs none	**0.0024**	1.000	1.003	1.000	1.000	1.000	1.002	1.02	**1.004**	1.000	1.004	1.000
quadratic trend vs none (log)	–	0.89	1.25	–	1.004	1.001	**0.50**	**1.18**	1.04	**0.82**	–	**0.76**
sparse PCE vs GP	**0.002**	**1.23**	**3.83**	1.02	**1.78**	**1.59**	**1.45**	**2.36**	**3.35**	**1.25**	**3.64**	**1.39**
sparse PCE vs GP (log)	–	1.10	**3.48**	–	**3.17**	**2.07**	**0.72**	**2.91**	**11.66**	**1.43**	–	**2.35**
committee vs GP + trend	0.995	0.96	**2.36**	1.02	**1.67**	1.16	**1.55**	1.41	**2.21**	1.002	**1.90**	1.08
CV weights vs uniform	**0.00047**	0.98	**0.84**	0.991	**0.54**	**0.86**	**0.60**	**0.55**	0.82	0.994	**0.38**	0.96
adaptive vs passive GP	1.007	1.18	0.94	1.06	0.86	1.01	1.02	1.07	0.96	1.22	1.06	0.93
adaptive vs passive committee	**0.80**	**1.25**	**0.61**	0.98	**0.70**	0.84	**0.69**	**0.60**	–	–	–	**0.84**
UEigV vs variance	1.02	0.90	**1.67**	1.07	**1.72**	**1.26**	–	–	–	–	–	–
AK-MCS vs passive, exceedance	**0.47**	0.88	**0.11**	0.88	**0.29**	0.31	**0.26**	0.39	**0.22**	1.41	0.35	–
AK-MCS vs passive, exceedance (log)	–	1.13	0.78	–	0.65	**0.17**	0.52	0.44	**0.27**	**0.35**	–	0.94
AK-MCS at $y_{0.95}$ vs passive	–	1.13	–	–	–	–	–	–	–	0.61	–	–
AK-MCS at $y_{0.95}$ vs passive (log)	–	0.68	–	–	–	–	–	–	–	0.55	–	0.95
log vs raw, exceedance	–	0.39	**0.21**	–	**0.59**	0.85	**0.39**	0.50	**0.19**	0.43	–	0.50

5 Results

Table 2 gives the median nRMSE of every strategy on every model at $N = 20$, Table 3 the paired comparisons and Fig. 2 the same relative to the Matérn ARD GP with confidence intervals.

5.1 GP specification and variable transformation

Estimating the kernel amplitude and one length scale per input (Matérn ARD versus the original kernel) reduced the nRMSE on eight of the eleven analytical models, by factors of 0.14 (physics) to 0.71 (piston), and did not change it on the plume (both input models) and Ishigami models, where neither kernel can resolve the response with 20 runs. On the physics model the original kernel, with its amplitude fixed at the variance of the standardised output, cannot represent a quadratic bowl whose values far exceed one standard deviation and reverts towards the mean between runs. Estimating the amplitude alone reduced the median nRMSE from 0.59 to 0.033, eighteen-fold; the earlier analysis had attributed this error to the size of the input domain (Section 6.1).

Fitting on the log scale, with log-transformed positive inputs, reduced the nRMSE on all eight analytical models with positive outputs, by factors of 0.11 (borehole) to 0.75 (piston); all eight reductions remain significant after the Holm adjustment. The GP then interpolates a nearly additive function of the log inputs, which is what products and power laws such as the plume, borehole and OTL formulas become.

Adding a quadratic trend to the GP reduced the nRMSE on the physics model by a factor of 400, to the level of the sparse PCE, and changed it by less than 0.5% on nine of the other ten models. The exception is the OTL circuit with 20 inputs, where the trend made the GP 1.6 to 8 times worse in 12 of the 30 designs: with 20 inputs the quadratic kernel adds 231 regression terms, which 20 runs cannot constrain. On the log scale, the trend improved the plume models (0.42 to 0.35 with uniform inputs, 0.37 to 0.28 with GUM-type inputs), whose log concentration is close to quadratic in the inputs, and the wing weight model (ratio 0.50), and again made the 20-input model worse (1.18). A trend is therefore a candidate to be tested by the selection step, not a default.

Independent check of the implementations. On the six models with uniform inputs, our sparse PCE matched that of OpenTURNS [50] on the same designs (median ratio of nRMSE 0.98–1.00, equivalent

within ±20% on all six), and the Matérn GP matched OpenTURNS' GP on five of them (0.95–1.00). On the physics model the OpenTURNS GP was 9.2 times more accurate. The cause is the upper end of the amplitude search range, 10^3 in standardised output units, which the estimated amplitude reached in nearly every design on that model; among all models and specifications it was reached elsewhere only by the log-scale GP on the CFD case (3 of 6 designs checked), where widening the range changed the median nRMSE from 0.080 to 0.077. With the range widened to 10^6, the median nRMSE on the physics model fell from 0.033 to 0.00097 for the GP with estimated amplitude, from 0.036 to 0.0022 with ARD and from 0.079 to 0.013 for the Matérn ARD GP. The pre-specified range thus understates the gain from estimating the amplitude on this model, about 600-fold rather than eighteen-fold, which strengthens the conclusion that the earlier failure was an artefact of restricting this hyperparameter; the GP with a quadratic trend (0.00018) remains the most accurate GP there. Universal kriging with a full quadratic basis (OpenTURNS; $1 + d + d(d+1)/2$ coefficients estimated by generalised least squares) was 1.3–3.3 times less accurate than the kernel-form trend used here on five of the six models, and PC-Kriging [51], the GP with the selected sparse PCE basis as trend, was not more accurate than the candidate chosen by LOO error on any of them.

5.2 Sparse PCE and committees

Sparse PCE matched the trend GP on the physics model and the Matérn GP on the Ishigami model, and was 1.2 to 3.8 times worse than the GP on the other nine models; with 20 runs, the corrected LOO error selects low-degree expansions that cannot follow the borehole, OTL and flood responses as closely as the GP. On the log scale it was 1.1 to 12 times worse, with one exception: on the wing weight model, a product of power laws that is nearly linear in the log inputs, the log-scale sparse PCE was better than the log-scale GP (ratio 0.72).

The CV-weighted committee was never better than the GP with a quadratic trend on the same design. It tied on six models and was 1.5 to 2.4 times worse on the other five: the borehole (both input models), OTL, wing weight and flood models. Its advantage on the physics model in the earlier analysis is entirely that of its degree-2 PCE member, which the trend GP reproduces (ratio 1.00). CV weighting remained necessary for a committee: it was never worse than uniform averaging and better on seven models, by up to three orders of magnitude on the physics model, where uniform averaging dilutes the exact member with poor ones.

5.3 Adaptive and targeted designs

Space-filling adaptive designs gave no consistent gain at $N = 20$. Adding runs at the largest GP posterior SD never changed the nRMSE significantly (ratios 0.86–1.22). The adaptive committee helped on the physics, borehole, OTL, wing weight and 20-input OTL models (0.60–0.80) and hurt on the plume model (1.25), where its estimate of $P(C > \text{IDLH})$ was off by a median of 0.17, against 0.02 for the passive committee. The spectral criterion UEigV was never better than plain committee variance and was worse on three models (1.26–1.72).

Targeting the threshold changes the picture for exceedance probabilities. AK-MCS reduced the median absolute error of $P(Y > T)$ significantly in 8 of the 19 comparisons, by factors of 0.11 to 0.47, and was never significantly worse. The largest gains were on the piston model (0.0099 to 0.0014 on the log scale), the borehole model (0.0091 to 0.0009 on the raw scale) and the GUM-type input models. Targeting comes at a cost elsewhere: on the Ishigami model the AK-MCS design raised the median error of the propagated mean from 8.5% to 28%, because its runs cluster near the threshold. Where the physical threshold lies in the bulk of the distribution (the IDLH on the plume models and the CFD

case), targeting the reference 95th percentile instead reduced the error of $P(Y > y_{0.95})$ on the log scale by median factors of 0.68 (plume) and 0.55 (plume with GUM-type inputs), but neither reduction survived the Holm adjustment, and on the CFD case it made no difference (0.95).

What the comparisons without a difference show. Of the 228 paired comparisons, 105 showed no significant difference after the Holm adjustment. Of these, 55 were equivalent within ±20% and 24 within ±10%; the rest are inconclusive with 30 seeds. The statements of no gain above are equivalences where they concern the committee against the GP with trend (6 of its 7 ties), the quadratic trend on responses without one (9 of 10) and the kernel specification on the plume (both input models), Ishigami and CFD models (4 of 4), but only in part for the GP-SD design (14 of 27), whose effect varies too much between designs for 30 seeds to show either a gain or its absence in the other 13 cases. All 12 non-significant comparisons of AK-MCS are inconclusive rather than equivalent.

5.4 Propagated quantities

Table 4 gives the validation errors of the reported quantities for the GP recommended for each model and for the committee. With the recommended GP, the 95th percentile was within 0.2% (OTL, flood) to 16% (Ishigami) of the reference, and $P(Y > T)$ within 0.0013 to 0.036 of its reference value. The committee's 95th percentile was off by 6.3–31% on seven of the eleven models. At the IDLH threshold of the plume model, the log-variable GP estimated $P(C > 210\,\mathrm{mg\,m^{-3}}) = 0.123$ with a median absolute error of 0.009, against 0.023 for the raw-variable GP.

5.5 Selecting the surrogate from the runs alone

The procedure of Fig. 6 fits the candidates (Matérn GP, GP with quadratic trend and sparse PCE, and their log-scale versions for positive outputs) and keeps the one with the smallest LOO RMSE, computed on the original scale of Y from the N runs alone. Applied to every seed, this choice had a median nRMSE between 1.000 and 1.044 times that of the best candidate in hindsight on the eleven analytical models, and was the best candidate itself in 27–77% of seeds; the largest shortfalls were on the Ishigami and plume models, where several candidates are almost equally poor or equally good. Relative to always using the Matérn GP, the selection reduced the nRMSE by a factor of 500 on the physics model (where it chose the sparse PCE or the trend GP), by factors of 0.12–0.84 on the models with positive outputs (choosing a log-scale GP in most seeds), and left it unchanged on the flood and Ishigami models. On the 20-input model it never chose the trend GP that overfits there, and on the wing weight model it chose the log-scale sparse PCE in 9 of 30 seeds. The selection step therefore recovers practically all of the benefit of the right specification without access to the true model.

5.6 Dependence on the number of runs

Fig. 1 repeats the main strategies with $N = 10$, 40 and 80 runs on the eight models with uniform inputs. The ranking of the specifications is stable: the original kernel stays the worst GP at every budget, and the benefit of the log scale grows with N, from factors of 0.62–1.00 at $N = 10$ to 0.10–0.48 at $N = 80$ on the six positive models, because the transformed response is smooth enough to be learned to high accuracy once the runs resolve it. Sparse PCE remains best on the physics model at every budget (at $N = 10$ it already has an nRMSE of 1.6×10^{-4}, against 0.97 for the Matérn GP) and worse than the GP on most of the others.

The verdict on space-filling adaptive designs depends on the budget. At $N = 10$ and 20, adding runs at the largest GP posterior SD never helped significantly. At $N = 40$ and 80 it reduced the nRMSE on

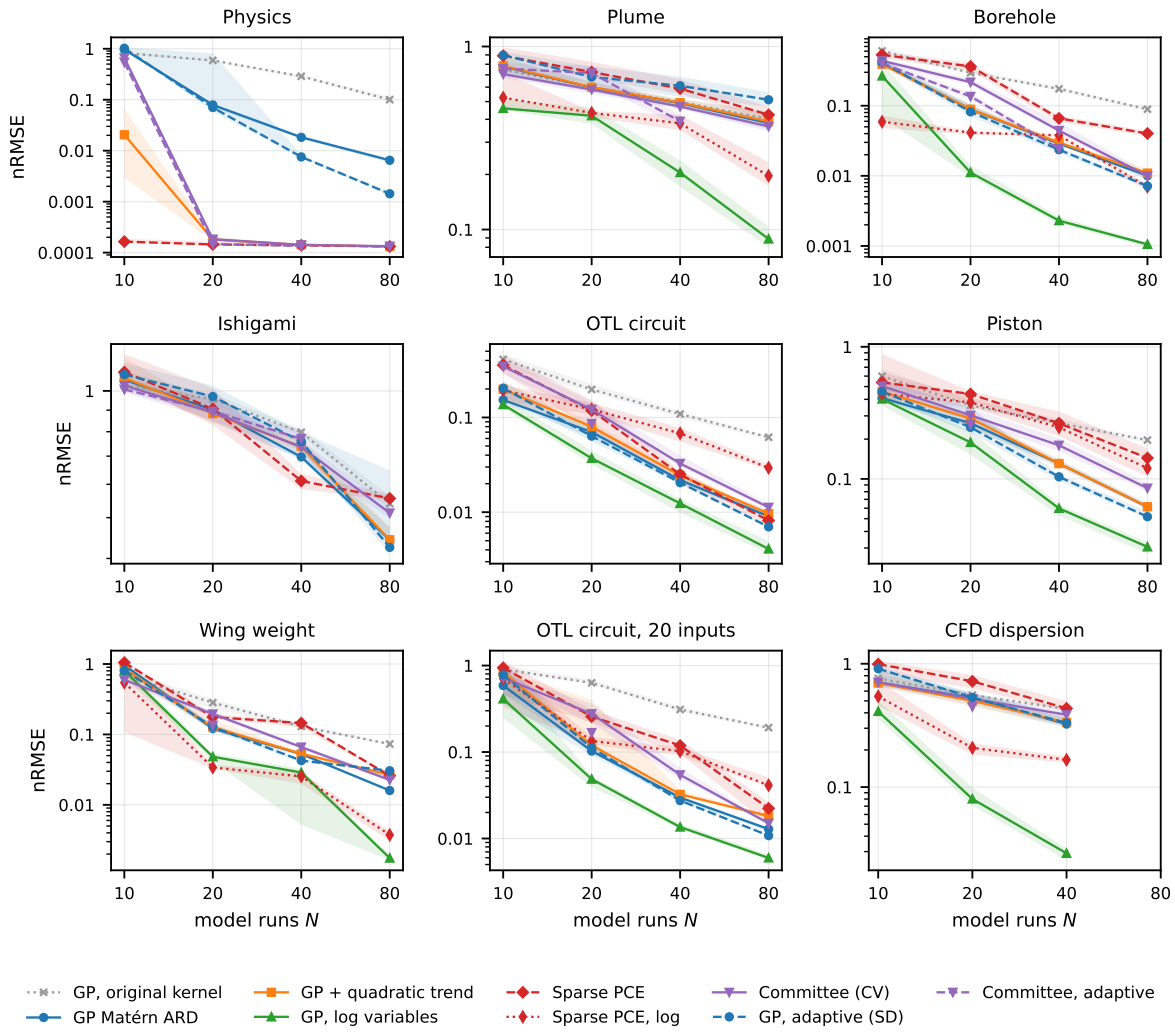


Figure 1: Median nRMSE over 30 seeds against the number of runs N for the eight models with uniform inputs; shaded: 95% bootstrap confidence interval of the median. Adaptive committees were run up to $N = 40$.

the physics (0.22–0.43), borehole (0.77), OTL (0.75 at $N = 80$), piston (0.81–0.87), wing weight (0.81 at $N = 40$) and 20-input OTL (0.83 at $N = 80$) models, which are smooth and on which the GP's uncertainty is informative once its hyperparameters are well determined, but increased it on the plume model (1.19 at $N = 40$), whose error concentrates in the high-concentration corner that the posterior SD does not single out, and on the wing weight model at $N = 80$ (1.93). The adaptive committee was better than the passive one at $N = 40$ on the physics and borehole models (0.94 and 0.50), no different elsewhere, and no better at $N = 10$. Adaptive selection is thus worth considering when the budget is several times the number of inputs, but not as a remedy for very small budgets, when the surrogate's own uncertainty is itself poorly estimated.

5.7 Is the surrogate contribution valid?

Fig. 3 and Table 5 test the three models of the surrogate error of Section 2.2 against the true models, for every surrogate fitted to 20 runs (68 combinations of model and surrogate on the eleven analytical models, 30 seeds each).

Pointwise intervals. The jackknife+ interval covered between 95.2% and 99.6% of the test points (median 97.4%) in every combination, in line with its distribution-free guarantee of at least 90%. The

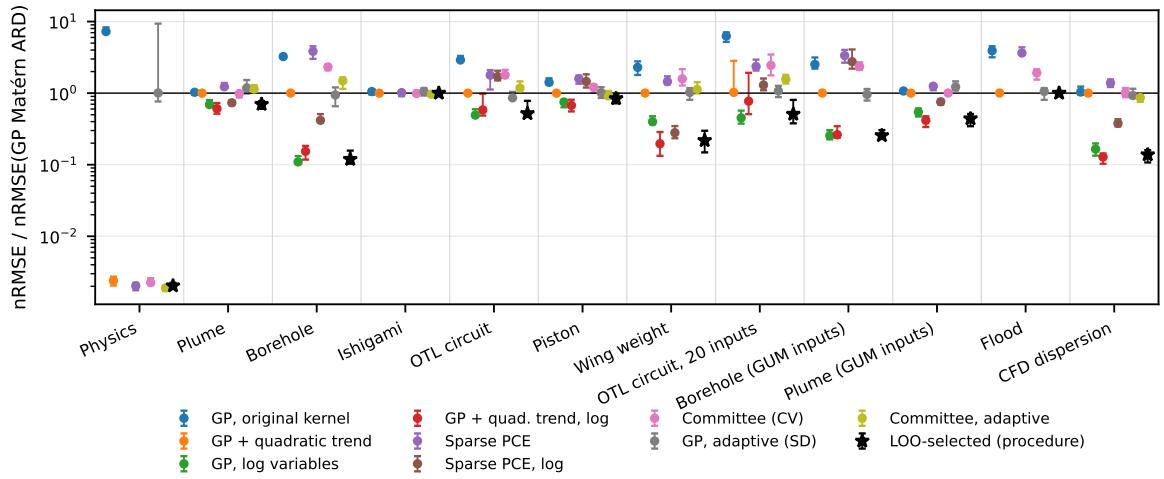


Figure 2: nRMSE of each strategy relative to the Matérn ARD GP on the same design, $N = 20$: median over 30 seeds of the paired ratio with 95% bootstrap interval. Star: the candidate chosen per seed by the smallest LOO error (the procedure of Fig. 6).

CV-normal interval covered 89.7–97.1% (median 94.2%). The GP's own predictive interval covered 68–98% (median 88%) and fell below 80% on the plume (both input models), Ishigami, piston, wing weight and 20-input OTL models: its posterior SD reflects the fitted covariance, not the actual error. Validity has a price in width: jackknife+ intervals were a median 1.6 (1.1–2.9) times as wide as CV-normal ones, and the narrow GP intervals still had the best interval score [49] in 31 of the 49 GP combinations. For a measurement uncertainty statement, which claims a coverage probability, the coverage is the property that matters.

GUM S1 coverage interval. Treating the surrogate as exact gave coverage intervals for Y whose achieved coverage under the reference distribution ranged from 57% to 96% (median 94%), and standard uncertainties $u(y)$ that were a median 0.95 times the reference (down to 0.41). Adding the LOO error to every Monte Carlo trial, equation (1), raised the achieved coverage to 91.5–97.3% (median 94.9%) as a normal error and to 91.0–97.0% (median 94.9%) by resampling the residuals, with $u(y)$ a median 1.00 (0.78–1.29) times the reference. The GP's predictive SD gave 89.8–96.0% (median 94.6%). The largest correction occurred where it is needed most: for the GP with the original kernel on the Ishigami model, coverage rose from 57% to 95%. Adding the error did not systematically improve exceedance probabilities, which depend on the tail of the surrogate rather than on its overall spread.

Reported quantities. The difference between the approaches is largest for the quantities a GUM S1 evaluation reports. Intervals for $y_{0.95}$ from joint GP posterior sample paths contained the true value in a median 63% of seeds (7–100%), and those for $P(Y > T)$ in 63%: the GP posterior describes the uncertainty of the interpolant as the fitted covariance model sees it, and is overconfident wherever that model is misspecified. The jackknife intervals contained the true value in a median 97% of seeds for both quantities, with the lowest values for the original-kernel GP, whose error is a bias that no resampling of its own fits can reveal.

5.8 Robustness of the surrogate contribution

Hyperparameter uncertainty. Integrating out the GP's hyperparameters narrowed the gap but did not close it (Table 6). Over the 42 combinations of model and GP candidate at $N = 20$, the median pointwise

Table 4: Validation error of the reported quantities, median over 30 seeds, $N = 20$: relative error (%) of the mean, SD and $y_{0.95}$ and absolute error ($\times 100$) of $P(Y > T)$ for the recommended GP (log scale for positive outputs), the same with an AK-MCS design, and the CV-weighted committee. Generated from the result records.

	GP (passive)				AK-MCS	Committee	
Model	mean	SD	$y_{0.95}$	P	P	$y_{0.95}$	P
Physics	1.36	4.22	3.88	1.37	0.77	0.41	0.15
Plume	4.74	15.9	6.94	0.91	1.13	23.1	2.36
Borehole	0.17	0.58	0.60	0.18	0.13	10.9	3.05
Ishigami	8.48	18.8	16.0	3.61	3.19	28.7	2.89
OTL circuit	0.17	0.60	0.21	0.14	0.14	2.29	1.31
Piston	0.62	2.75	1.91	0.99	0.14	7.47	2.68
Wing weight	0.07	1.03	0.45	0.30	0.15	3.87	1.89
OTL circuit, 20 inputs	0.14	0.64	0.32	0.21	0.09	6.31	2.13
Borehole (GUM inputs)	0.09	1.22	0.47	0.13	0.04	6.85	2.25
Plume (GUM inputs)	6.27	17.0	8.76	0.80	0.29	31.2	3.72
Flood	0.06	1.81	0.18	0.41	0.17	0.75	1.16
CFD dispersion	4.41	7.62	6.89	0.85	0.90	27.1	5.86

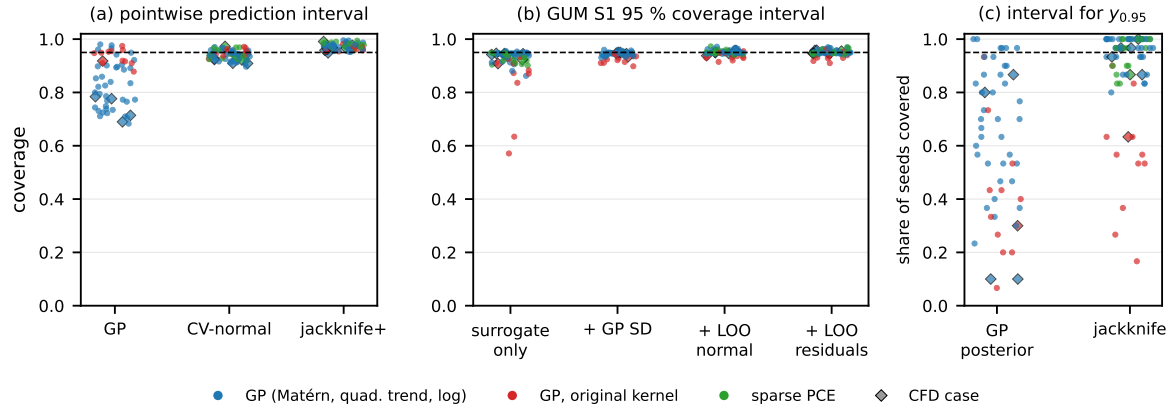


Figure 3: Coverage of nominal 95% intervals, mean over 30 seeds; one marker per combination of model and surrogate (circles: the eleven analytical models; diamonds: the CFD case, Section 5.9). (a) Pointwise prediction intervals on 5000 test points. (b) Achieved coverage of the GUM S1 95% coverage interval of Y under the reference distribution, with the surrogate treated as exact or with its error added to each trial. (c) Share of seeds whose 95% interval for $y_{0.95}$ contains the true value. Dashed line: 0.95.

coverage rose from 80% with the plug-in hyperparameters to 83% for the Student-t process and 96% for the fully Bayesian GP; the recalibrated GP reached 95% (91–98%) and the jackknife+ 97%. For the reported quantities the gain was smaller. Intervals for $y_{0.95}$ contained the true value in at least 26 of the 30 seeds, the threshold fixed in advance, in 13 of the 42 combinations with the plug-in GP, 15 with the Student-t process, 24 with the fully Bayesian GP, 23 with the recalibrated GP and 39 with the jackknife. The fully Bayesian GP fell as low as 8 of 30 seeds (Matérn GP on the CFD case): averaging over hyperparameters cannot repair a covariance model that does not fit the response, here the raw scale of a positive, skewed output. The jackknife fell short of the threshold on three combinations, all raw-scale GPs on the plume model with GUM-type inputs and on the Ishigami model (24–25 of 30). By the rule fixed in advance, none of the GP-based descriptions is a valid surrogate contribution on all models, and the recommendation of the LOO residuals stands. The coverage interval of Y itself was robust to the choice: 93% or more with every error model. The MCMC chains had integrated autocorrelation times of 38–62 steps, so the 500 steps kept per walker span 8–13 autocorrelation times.

Table 5: Coverage (%) of nominal 95% intervals for the GP recommended for each model (log variables for positive outputs), 30 seeds, $N = 20$. Pointwise: share of test points. GUM S1: achieved coverage of the 95% coverage interval of Y. $y_{0.95}$, $P(Y > T)$: share of seeds whose interval contains the true value (GP: joint posterior paths; J: jackknife). Generated from the result records.

	Pointwise			GUM S1 interval of Y				$y_{0.95}$		$P(Y > T)$	
Model	GP	CV	J+	exact	+LOO N	+LOO res.	+GP	GP	J	GP	J
Physics	98	96	97	92	95	95	94	90	87	87	70
Plume (log)	71	91	97	94	95	95	94	77	93	70	97
Borehole (log)	97	96	97	95	95	95	95	97	100	97	97
Ishigami	71	92	97	86	97	97	93	73	83	73	73
OTL circuit (log)	90	95	98	95	95	95	95	93	100	93	100
Piston (log)	77	91	95	94	94	94	94	63	93	60	90
Wing weight (log)	79	94	98	95	95	95	95	70	97	93	100
OTL circuit, 20 inputs (log)	92	97	99	91	95	95	95	97	100	93	100
Borehole (GUM inputs) (log)	96	96	97	95	95	95	95	97	100	97	100
Plume (GUM inputs) (log)	73	93	98	94	95	95	94	60	100	57	100
Flood	90	95	98	95	95	95	95	97	100	73	100
CFD dispersion (log)	78	91	95	95	95	95	95	80	97	87	97

Number of runs. The procedure (the candidate with the smallest LOO error in each seed, with its LOO residuals added) gave GUM S1 coverage intervals that covered 93.4–96.9% of the reference distribution at $N = 10$, 94.2–96.1% at 20, 94.7–95.7% at 40 and 94.7–95.1% at 80, on every model (Table 7). Treating the chosen surrogate as exact covered as little as 81.9% at $N = 10$, against 89.9% at 20: the contribution matters most when the budget is smallest. With ten runs, a design's coverage fell below 90% in up to 23% of the designs of a model, with 20 runs in at most 3% and with 40 or 80 runs in none. The selection was within 4.4% of the best candidate in hindsight at every budget from 20 runs, but up to 37% worse at $N = 10$, where ten LOO residuals rank the candidates less reliably. A 95% jackknife+ interval needs at least 19 runs; with ten it is unbounded. The jackknife intervals for $y_{0.95}$ contained the true value in 19–30 of 30 seeds at $N = 10$ and 23–30 at $N = 40$. At $N = 80$ they did so in 14 of 30 seeds on the physics model, whose surrogate error at that budget (a relative error of $y_{0.95}$ below 10^{-4}) is at the level of the observation noise shared by all LOO fits, which no resampling of the fits can reveal.

Single designs. A practitioner has one design. At $N = 20$, with the LOO residuals added, the achieved coverage of the GUM S1 interval was below 90% in at most 10% of the designs of any model and surrogate (median 0%), and never below 80.8%.

A feature the runs miss. On the bump model, 13 of the 30 designs of 20 runs placed a run inside the bump. When the design missed it, the GUM S1 interval still covered 94–95% of the reference distribution, because the bump holds only 3% of the probability mass, and the jackknife+ intervals 95–97% of the test points; but the intervals for $y_{0.95}$ contained the true value in only 9–10 of the 17 designs, against 13 of 13 when the bump was hit. With 40 runs, 6 designs missed the bump, and their intervals for $y_{0.95}$ contained the true value in 1–4 of 6. Nothing computed from the runs can flag this: the LOO residuals, like every estimate from the runs, describe only the errors that the runs reveal. A local feature smaller than the spacing of the design needs prior knowledge of where to look, or runs placed there.

Several output quantities. For the plume model at three receptors, the LOO residuals of the three components were strongly correlated (median correlation 0.83 over the 30 designs). Without the surrogate error, the GUM S2 hyperellipsoidal region covered 93.4% of the reference distribution on the original scale and 92.4% on the log scale. With the residual vectors resampled jointly it covered 95.6% and 96.9%, with a volume 2.7 and 2.6 times that of the region computed from the true model. Resampling the components independently, or independent normal errors, covered 96.7–98.4% but with regions 5.7–7.2 times the reference volume: ignoring the correlation of the surrogate errors over-states the joint uncertainty.

Table 6: GP-based descriptions of the surrogate error against the jackknife, $N = 20$, for the GP with a quadratic trend on the recommended scale (log for positive outputs). Pointwise: mean coverage (%) of 95% prediction intervals on the test points. $y_{0.95}$: seeds (of 30) whose 95% interval for the 95th percentile contains the true value. Plug-in: ML hyperparameters; t: Student-t process; MCMC: all hyperparameters integrated out; recal.: GP SD recalibrated by the LOO standardised residuals; J+/J: jackknife+ and jackknife. Generated from the result records.

	Pointwise coverage (%)					$y_{0.95}$: seeds of 30 covered				
Model	plug-in	t	MCMC	recal.	J+	plug-in	t	MCMC	recal.	J
Physics	89	91	96	94	97	21	21	29	25	30
Plume	77	79	92	94	96	17	20	29	28	29
Borehole	98	98	99	97	97	30	30	30	30	30
Ishigami	70	73	93	91	97	20	23	26	15	29
OTL circuit	82	85	96	92	98	26	26	30	28	30
Piston	80	83	93	94	95	19	21	24	24	28
Wing weight	83	85	99	97	99	25	25	30	28	30
OTL circuit, 20 inputs	95	96	97	97	99	27	29	29	28	30
Borehole (GUM inputs)	94	95	99	95	97	28	28	30	28	29
Plume (GUM inputs)	79	81	93	96	97	21	23	28	29	30
Flood	90	92	98	96	99	30	30	30	29	30
CFD dispersion	78	82	98	93	98	26	27	30	27	29

Table 7: Achieved coverage (%) of the GUM S1 95% coverage interval of Y for the procedure (candidate chosen by LOO error, LOO residuals added), mean over 30 seeds, against the number of runs. The CFD case was not run with 80 runs. Generated from the result records.

Model	$N = 10$	20	40	80
Physics	95.6	95.0	95.0	95.0
Plume	94.0	94.7	94.8	94.8
Borehole	93.4	94.9	94.9	94.9
Ishigami	96.0	96.1	95.7	94.7
OTL circuit	95.6	94.9	95.0	95.0
Piston	94.0	94.2	94.9	95.1
Wing weight	94.6	94.6	94.8	94.8
OTL circuit, 20 inputs	95.4	94.7	94.8	94.9
Borehole (GUM inputs)	94.7	94.6	94.8	94.9
Plume (GUM inputs)	94.9	95.1	94.9	94.9
Flood	96.9	94.6	94.9	94.9
CFD dispersion	94.0	94.6	94.7	–

5.9 CFD dispersion case

The case. Fig. 4 shows the flow and concentration field for the centre of the input box: the plume is lifted over the building, part of it is trapped in a recirculating wake that extends several building heights downstream, and the concentration at breathing height is nearly uniform inside the wake and decays beyond it. Of the 2900 CFD runs of the study (1550 for the designs and the reference, the rest added by the adaptive designs), 80% met the residual criteria within 4000 iterations (median 1322 iterations); for the others, mostly tall buildings whose steady wake keeps the residuals oscillating, the receptor concentration varied by less than 0.16% over the last 2000 iterations, so the iteration error is negligible. Refining the mesh twice in each direction changed the receptor concentration by 0.9–6.5% (median 3.7%) in six flow states spread over the input box, and a mesh with half the cells in each direction gave values between 8.1% lower and 0.5% higher. The convergence was oscillatory at 21 of the 24 combinations of flow state and receptor distance, the standard mesh giving the largest value, and at the other three the differences grew with refinement. The solutions are therefore not in the asymptotic range, where the grid convergence index applies [52]; estimated from the range of the three solutions, the discretisation error of the standard mesh is about 6% (median 5.9%, at most 8.1%). It is part of the model as defined, as

it would be for any production CFD model. (The coarse-mesh runs reached the iteration limit without meeting the residual criterion of the scalar, but their receptor values changed by less than 4×10^{-5} after 1000 iterations.) The reference gives a mean of 434 mg m^{-3}, an SD of 413 mg m^{-3}, a 95th percentile of 1252 mg m^{-3} and $P(C > \text{IDLH}) = 0.673$, with standard uncertainties from a bootstrap over the 800 reference runs of 2.3%, 3.4%, 3.3% and 0.007; validation errors of the propagated quantities below about 5% are therefore not resolved.

Specification and selection. The CFD case confirms the analytical results (Table 2, last row). Kernel specification did not matter (ratio 0.96), but the log scale did: the log-variable GP had a median nRMSE of 0.080 against 0.53 for the Matérn GP (ratio 0.17, better in all 30 seeds), and a quadratic trend on the log scale reduced it further to 0.060 (ratio 0.76). Sparse PCE was 1.4 times worse than the GP on the raw scale and 2.3 times on the log scale, and the CV-weighted committee was no better than the GP with trend (ratio 1.08). The selection by LOO error chose a log-scale GP in all 30 seeds and was within 0.1% of the best candidate in hindsight. With the log-variable GP, the 95th percentile was within 6.9% of the reference and $P(C > \text{IDLH})$ within 0.009, against 18.5% and 0.019 for the Matérn GP and 27% and 0.059 for the committee.

With 10 and 40 runs the picture is the same: the log scale reduced the nRMSE of the Matérn GP by factors of 0.58 and 0.094, the GP-SD design made no difference (1.02 and 1.10) and sparse PCE remained less accurate than the GP (1.21 and 1.45).

Adaptive designs. Adding runs at the largest GP posterior SD did not change the nRMSE (ratio 0.93), and AK-MCS did not improve $P(C > \text{IDLH})$ (ratio 0.94): the IDLH is exceeded in two thirds of the cases, so the threshold lies in the bulk of the distribution rather than in a tail, where a targeted design has most to gain. The adaptive committee reduced the nRMSE relative to the passive one (ratio 0.84) but, as on the plume model, its estimate of $P(C > \text{IDLH})$ was off by a median of 0.11, against 0.06 for the passive committee and 0.009 for the log-variable GP.

Surrogate contribution. The coverage results carry over to the CFD case (Table 5, last row; diamonds in Fig. 3). Pointwise, the GP intervals covered 69–92% of test points and the jackknife+ intervals 94.8–99.1%. Treating the surrogate as exact gave GUM S1 coverage intervals with an achieved coverage of 90.8–94.7%; adding the LOO error gave 93.8–95.7%. The GP posterior was most misleading for the 95th percentile: with the raw-scale GPs (Matérn, with or without trend) its 95% interval contained the true value in only 10% of seeds, against 87–93% for the jackknife; with the log-variable GP the figures were 80% and 97%.

6 Discussion

6.1 Relation to the earlier analysis

The earlier analysis made five claims. Three are withdrawn or revised. The catastrophic GP failure on large domains is a kernel-specification artefact: with the amplitude estimated, the GP no longer reverts to the mean (Fig. 5). The spectral acquisition criterion gives no advantage over plain committee variance on any of the six models on which it was tested, and is worse on three. The committee's advantage is not general but comes from a member that matches the dominant term, which a GP with a quadratic trend or a sparse PCE reproduces on its own. Two claims survive: aggregation must be error-weighted, and an adaptive committee can be more accurate than a passive one: it was on four of the seven models on which it was run, including the CFD case, although it remained less accurate than the log-variable GP wherever the output is positive, and its exceedance probabilities were worse on the plume and CFD cases. The records of the earlier campaigns are kept in the repository [31].

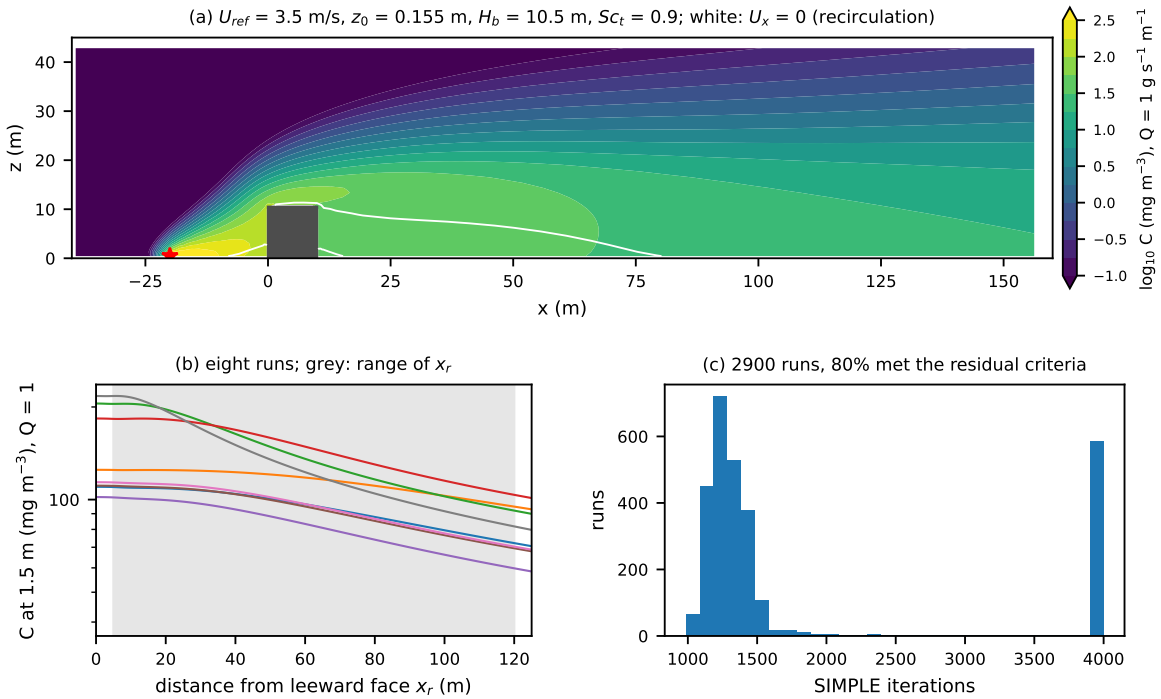


Figure 4: CFD dispersion case. (a) Concentration at a unit release rate for the centre of the input box; white line: zero streamwise velocity, enclosing the recirculating wake; star: release. (b) Concentration at 1.5 m height against distance from the leeward face for eight reference runs. (c) Iterations of the 2900 runs; the runs at 4000 did not meet the residual criteria but changed by less than 0.16% over their last 2000 iterations.

6.2 What the surrogate contribution should be

The coverage results separate two uses of a surrogate's uncertainty. Pointwise, the GP posterior is a useful but uncalibrated description: it locates where the surrogate is uncertain, which is why it drives good adaptive designs at larger budgets, but its scale is set by a covariance model fitted to 20 runs and is too small wherever that model is wrong. For an uncertainty statement the scale is what matters, and the leave-one-out residuals estimate it directly, without trusting the covariance model; the jackknife+ turns them into intervals with a guarantee. The difference is largest for the reported quantities. A GUM S1 coverage interval of Y is fairly robust, because surrogate error that varies across the input space partly averages out in the distribution of Y, and adding the LOO error to each trial restores its nominal coverage. A 95th percentile or an exceedance probability is not robust: it depends on the surrogate in a small region. Intervals from the GP posterior contained the true value in a median 63% of the seeds (median over model and surrogate), whereas the jackknife over LOO fits did so in a median 97%. Integrating out the hyperparameters narrowed this gap but did not close it (Section 5.8): the shortfall comes from covariance models that do not fit the response, not only from plug-in estimates, and a Bayesian treatment inherits the covariance model. Rescaling the GP SD by the LOO standardised residuals, which keeps the GP's localisation of the error and takes its scale from the runs, came closer, but the jackknife remained the more reliable description. We therefore recommend the LOO error as the surrogate contribution, and the jackknife over LOO fits for the uncertainty of quantities derived from the propagated distribution.

6.3 Limitations

The study covers twelve models with up to 20 inputs and budgets of 10 to 80 runs; problems with many more inputs, where sparse and additive structures matter more, may rank the surrogates differently. Two

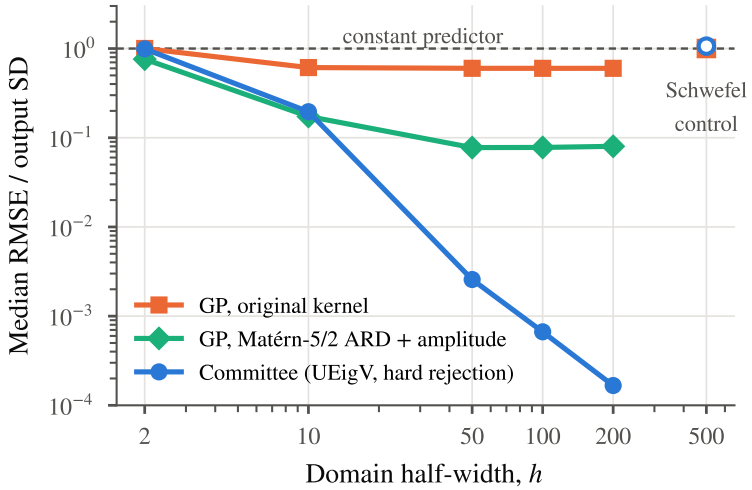


Figure 5: Physics model on $[-h, h]^3$, $N = 20$: median nRMSE over 30 seeds against the domain half-width, for the GP with the original kernel (amplitude fixed), the Matérn ARD GP with amplitude, and the committee of the earlier analysis. Open markers: Schwefel function on $[-500, 500]^3$. Dashed line: constant predictor.

models were run with a single budget per input distribution, and the adaptive committees were run only up to $N = 40$ because of their cost. The CFD case is two-dimensional and steady, with a standard k–ε model; it is a benchmark with the features a dispersion simulation has, not a validated prediction for a site, and its reference distribution is built from 800 runs rather than 2×10^5. The coverage results are averages over seeds; for single designs the GUM S1 interval fell below 90% in at most 10% of the designs of a model at $N = 20$, and the jackknife+ guarantee (at least 90% for nominal 95%) holds on average over designs, not for every design. The LOO residuals describe only the errors that the runs can reveal: a local feature that no run hits goes unnoticed (Section 5.8), and an error at the level of the observation noise, shared by all LOO fits, is not seen by the jackknife. The hyperparameter search ranges were fixed in advance; on the physics model the amplitude reached the upper end of its range, which understates the GP there (Section 5). Several output quantities were studied on one model only; there, resampling the LOO residual vectors run by run, which preserves their correlation, gave the nominal coverage of the GUM S2 region [41], and resampling the components independently over-stated the region. The log transformation applies only to positive outputs, and its back-transformation to the predictive median slightly underestimates the mean of a skewed output; the bias is included in the validation errors reported. Finally, the paired tests are exploratory: all p-values were Holm-adjusted together, and nominal differences that do not survive the adjustment are not claimed.

7 A procedure for surrogate-based GUM S1 evaluations

Fig. 6 condenses the results into a procedure that needs only the N runs. Its two key steps were validated on the twelve models: the selection of the surrogate by its LOO error recovered practically all the benefit of the best specification (Section 5.5), and the LOO error as surrogate contribution restored the nominal coverage of the GUM S1 coverage interval (Section 5.7). Candidates should include a GP with an estimated amplitude, one length scale per input and a quadratic trend, and a sparse PCE, each on the log scale when the output is positive; a fitted hyperparameter at the end of its search range is a sign that the range should be widened. With fewer than 19 runs a 95% jackknife+ interval is unbounded, but the GUM S1 coverage interval with the LOO residuals remained valid down to ten runs. The LOO error covers what the runs can reveal: where a narrow feature is plausible (a narrow plume, a local flow

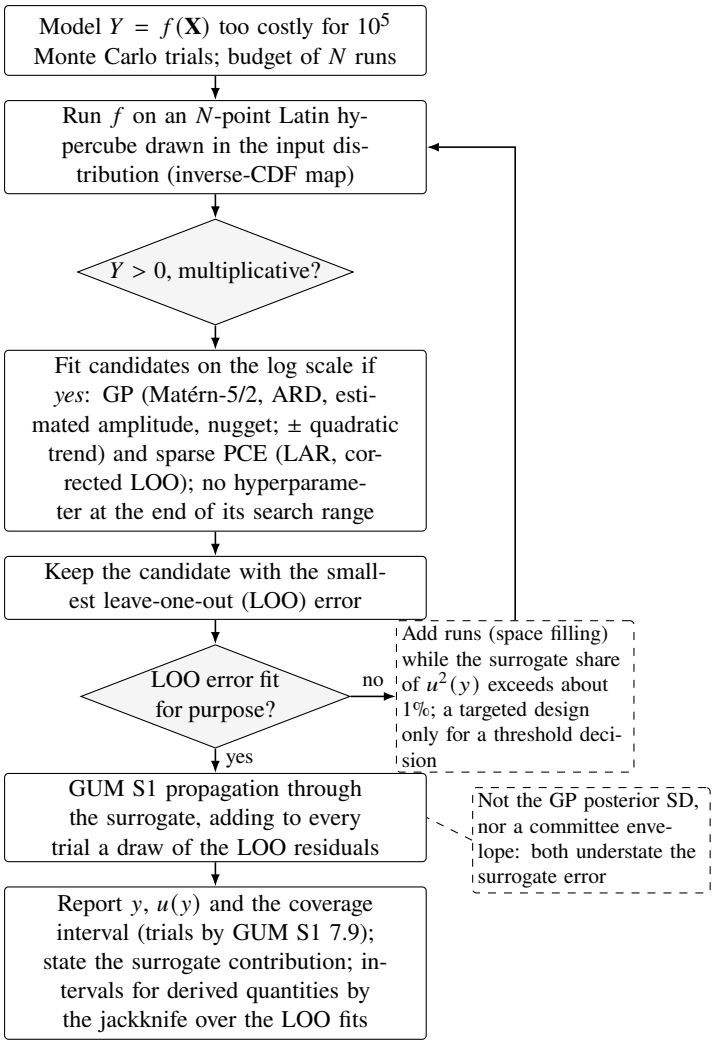


Figure 6: Procedure for a GUM S1 evaluation with a surrogate fitted to N runs of an expensive model.

structure), prior knowledge or runs placed there are needed. Adaptive designs are optional: a targeted design such as AK-MCS is worthwhile when a threshold decision is the purpose and the threshold lies in a tail of the distribution, and variance-driven designs when the budget is several times the number of inputs and the model is smooth.

When to stop adding runs. The procedure also indicates whether more runs are worthwhile. The share of the variance of Y due to the surrogate error, $1 - u_0^2/u^2$ with u_0 the standard uncertainty without and u with the LOO residuals added, is computed from the runs alone. Over the 7020 evaluations of the study (models, candidates, budgets and seeds) it ranked the actual surrogate error well (Spearman correlation 0.86 with the nRMSE). Where the share was below 1% (2512 evaluations), the median nRMSE was 0.026 and the standard uncertainty was within 0.6% (median) of the reference; for shares of 1–5%, 0.13 and 2.5%; above 5%, 0.51 and 16%. A share below about 1% therefore indicates that further runs would hardly change the uncertainty statement.

Cost. On one processor core, fitting and propagating a GP took 0.3–0.8 s and a sparse PCE 0.1 s; the LOO step of the procedure, every candidate refitted once per run, took 37 s (median, at most 74 s) with 20 runs; the committee took 9 s and the adaptive committee 78 s per design. One run of the CFD model, itself small, took 39 s on the same core. The cost of the procedure is therefore that of one or two runs of the model, and it falls further behind as the model becomes more expensive.

Table 8: Worked example: the procedure of Fig. 6 on the CFD case with $N = 20$ runs and $M = 1.15 \times 10^6$ trials, chosen by the adaptive procedure of GUM S1 (mg m^{-3}). Columns: surrogate treated as exact; with the LOO residuals added (the reported result); 95% jackknife interval over the LOO fits (surrogate error only); reference from 800 CFD runs with its standard uncertainty in parentheses.

Quantity	exact	with E	jackknife	reference
Estimate y (mean)	419	414	[401, 437]	434 (10)
Standard uncertainty $u(y)$	385	381	[359, 410]	413 (14)
95% coverage interval, lower	40	39	[38, 42]	39 (0.8)
95% coverage interval, upper	1471	1455	[1378, 1564]	1589 (59)
95th percentile $y_{0.95}$	1171	1158	[1108, 1235]	1252 (41)
$P(C >$ IDLH)	0.669	0.663	[0.651, 0.687]	0.673 (0.007)

7.1 Worked example

The procedure is implemented in `src/procedure.py` of the repository [31]; given the N runs and a sampler of the input distribution, it returns the GUM S1 results with and without the surrogate contribution, jackknife intervals of the reported quantities and an uncertainty budget. The number of Monte Carlo trials is chosen by the adaptive procedure of GUM S1 [2] (clause 7.9) for a stated number of significant digits of $u(y)$. Applied to the CFD case with the 20 runs of the first design, it chose the log-scale GP with a quadratic trend, whose LOO RMSE (23 mg m^{-3}) was smaller than those of the log-scale GP (31), the log-scale sparse PCE (93) and the raw-scale candidates (157–260). Two significant digits of $u(y)$ (numerical tolerance 5 mg m^{-3}) took 115 batches of 10^4 trials, $M = 1.15 \times 10^6$; the upper limit of the coverage interval was the last quantity to settle. Table 8 compares the result with the reference, which a practitioner would not have. Adding the LOO residuals changed the reported quantities by at most 1.3%; here it lowered them, because the residuals have a negative mean (-0.013 on the log scale, mostly from one run). The reported 95% coverage interval contains 94.3% of the reference distribution. The jackknife intervals, which describe the surrogate error alone, contain the reference values of the mean, the lower limit of the coverage interval and $P(C >$ IDLH); for the SD, the 95th percentile and the upper limit the reference lies 0.7–1.6% above the upper end of the interval, by less than half the standard uncertainty of the reference itself.

The GUM uncertainty framework, the law of propagation of uncertainty applied to the same surrogate at the input means, gives $y = 318$ mg m^{-3}, $u(y) = 232$ mg m^{-3} and the interval $y \pm 1.96\,u(y) = [-137, 772]$ mg m^{-3}, which contains only 85.5% of the reference distribution and extends to impossible negative concentrations. For this skewed output the Monte Carlo method of GUM S1, and therefore a surrogate, is needed.

The budget treats the surrogate error E as one more input quantity of the model (1), with E drawn from the LOO residuals, and gives first-order variance shares of the inputs and the total share of E (estimators of Saltelli et al. [53]). The reference wind speed contributed 36% of Var(Y), the release rate 29%, the receptor distance 5.9%, the turbulent Schmidt number 4.2%, the building height 3.2% and the roughness length 1.7%; interactions between the inputs 20% and the surrogate 0.3%. With this surrogate, then, the evaluation is limited by the input uncertainties rather than by the number of runs, and further CFD runs would mainly narrow the jackknife intervals of the tail quantities. The budget is how a practitioner can tell whether more runs are worth their cost: a surrogate share comparable to those of the inputs would indicate the opposite.

8 Conclusion

For GUM S1 evaluations in which the measurement model can be run only a few tens of times, the way a surrogate is specified matters more than the strategy around it. A GP should estimate its amplitude and one length scale per input and work on the log scale when the output is positive; a failure of GP surrogates reported in an earlier analysis was an artefact of one hyperparameter held fixed. A quadratic trend helps when the response has one but can overfit when there are many inputs, so it belongs among the candidates rather than in every model. Among such candidates and a sparse PCE, the one with the smallest leave-one-out error was as good as the best in hindsight on all twelve models, including a CFD simulation of a release behind a building. Committees of surrogates and variance-driven adaptive designs gave no consistent gain at 20 runs, whereas a design targeted at the threshold improved exceedance probabilities when the threshold lay in a tail of the distribution. Finally, the surrogate's own error must be part of the uncertainty statement, and it should be estimated from leave-one-out residuals rather than from the GP's predictive variance: the former restored the nominal coverage of the GUM S1 coverage interval and gave valid intervals for reported quantities, the latter was overconfident precisely where the surrogate was poor.

Data and code availability

All code, the per-seed records of every strategy, the cached CFD runs, figure scripts, an analysis script that recomputes every table and figure, and an implementation of the procedure of Section 7 (`src/procedure.py`) are available at `https://github.com/Yugam2508/fyp-active-learning-cfd` [31]; `docs/REPRODUCIBILITY.md` lists the commands and the software versions.

CRediT authorship contribution statement

Yugam Jinesh Vora: conceptualisation, methodology, software, investigation, formal analysis, visualisation, writing – original draft. Wee Hoe Ng: conceptualisation, supervision, writing – review and editing.

Declaration of competing interest

The authors declare no competing interests.

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